FIRST ROBOTICS PROJECT

ROBOTICS

POLITECNICO
MILANO 1863

The Project

Compute robot odometry

with partial data from encoders

publish pose as tf

compute error between GT and your odom

Data

Format: ROS Bag file

play the bag with the command:

ros2 bag play bag_name

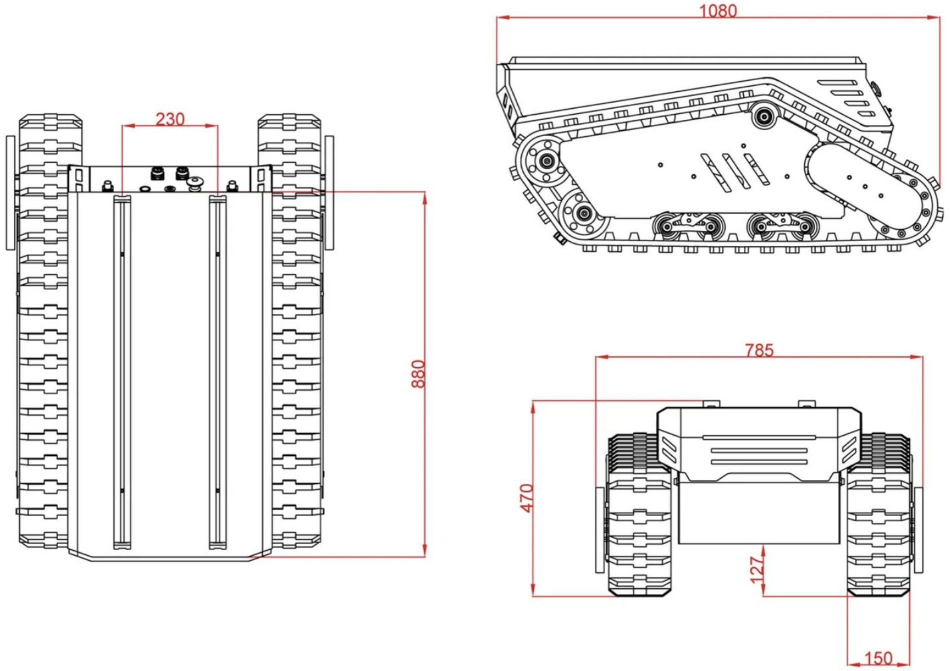
Data:

- /bunker_status robot data
- /odom gps front robot odometry
- /tf gps tf

The robot

Agile-x Bunker Pro

MAX SPEED	1.5M/S
MAX TRAVEL	15KM（FULL PAYLOAD）



The project (First Node)

- Create a ROS package called ***first_project***
- Create a ROS node to compute the odometry from robot data:
  - write a node called ***odometer*** which
    - subscribe to:
      - robot status:
        - type: ***bunker_msgs/msg/BunkerStatus*** (it's a custom msg...)
        - topic name: ***/bunker_status***
    - publish:
      - odometry message:
        - type: ***nav_msgs/Odometry***
        - topic name: /***project_odom***
      - **tf odom-base_link2**

The project (First Node)

- The node also implement a service called ***reset***
- When the service is called it resets the odometry value to zero on position and orientation

The project (Second Node)

- Create a ROS node to compute the error of the computed odometry:
  - write a node called **tf_error** which
    - compute the distance between the tf of the bag and your tf
    - this means the distance between base_link and base_link2
    - publish:
      - custom message:
        - type: ***first_project/tf_error_msg***
        - topic name: **/tf_error_msg**
        - fields:
          - header header
          - float32 tf_error
          - int time_from_start
          - float32 travelled_distance (in meters)

The project (Launch file)

- The two node should all start from a single launch file called **first_project.launch.py**
- The launch file should also open rviz loading a configuration file which shows the two tfs in top view with names for the tfs
- the only command to start the project should be:
  ros2 launch first_project first_project.launch.py

Deadlines and requested files

-**Only** a zip file is required as submission

-Upload on Webeep

-Inside the archive:

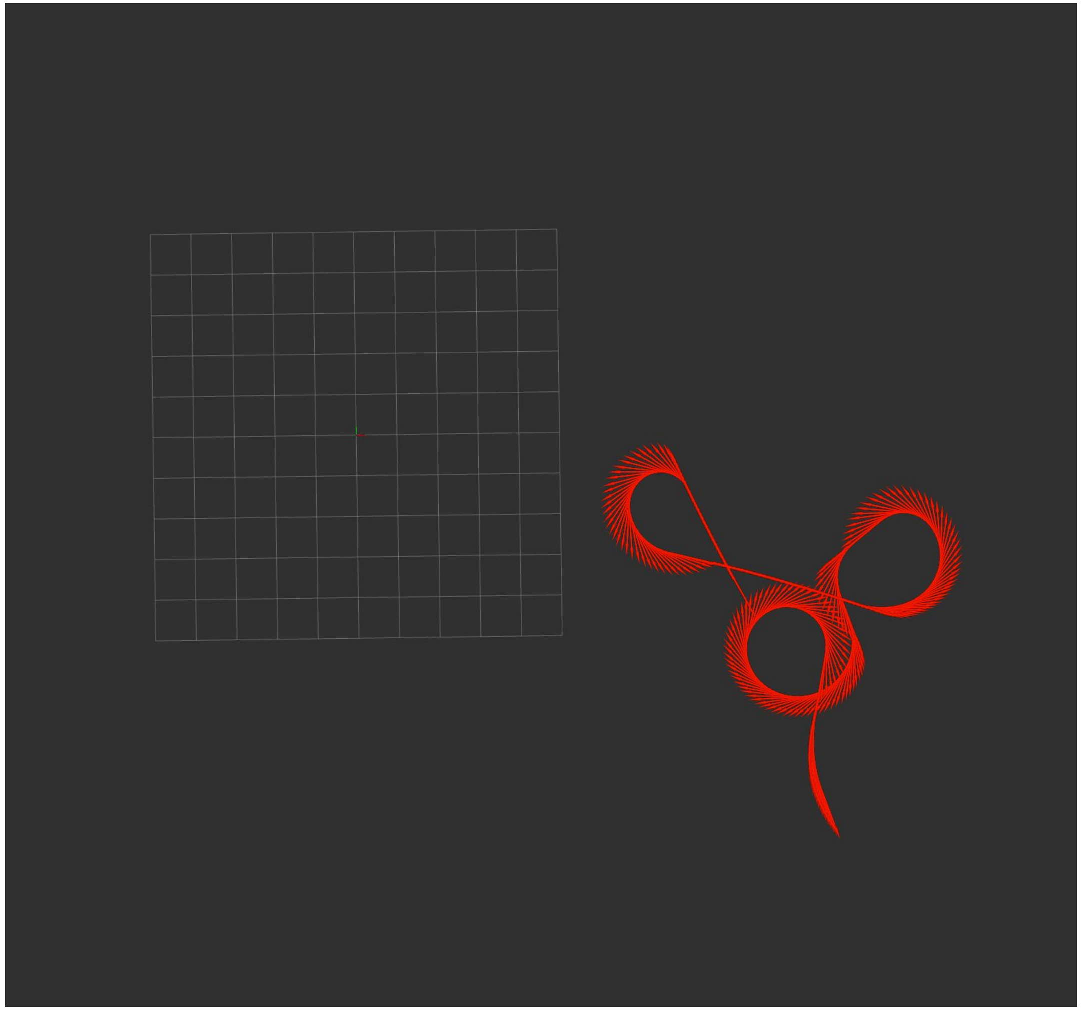
- info.txt file (details next slide)
- folders of the nodes you created (with inside CmakeLists.txt, package.xml, etc...)
- screenshot folder with the top view in rviz of the path of each bag, using your computed odometry with keep equal to 100000 (see next slide for example)
- **do not send** the entire environment (with build and devel folders)
- **do not send** the bag files

Deadlines and requested files

Example screenshot for one bag

To be included in the screenshot folder

Name the files with the bag name



Deadlines and requested files

File txt must contain only the group names with this structure

codice persona;name;surname

You can add another file called readme.txt with additional info. I will not always look for it. But if something goes wrong I'll check for explanations.

Some more requests

Name the archive with your codice persona

Don’t use absolute path

The project need to be written using c/c++

How to get less points

- The project do not compile -> 0 points
- The project has absolute paths -> 0 points
- The archive has build and devel folder -> -1 points
- The archive has bag files -> -1 points
- The project do not open rviz -> - 1 points
- The project do open rviz with wrong config -> - 1 points
- Two members upload the project -> ½ points
- Three members upload the project -> ⅓ points

Deadlines and requested files

Deadline: 29 April (3 weeks)

Max 3 student for team

Questions:

-write to me via mail (simone.mentasti@polimi.it)

- do not write only to Prof. Matteucci